

Contents

LO 1- Web and Mobile Application Design Technology

Contents	1
1 Task1 Part 1	2
1.1 Modern Web and Mobile Development Technologies and Frameworks	2
1.2 Evaluating the Impact of Common Development Technologies and Frameworks on Design, Functionality and Management.....	3
1.3 Impact of Website Design on Search Engine Results	4
1.4 Improving Website Ranking Using SEO	4
2 Task1 Part 2	6
2.1 Design Flexibility, Performance, Functionality, User Experience (UX) and User Interface (UI).....	6
2.2 Tools, Techniques and Languages Used to Design and Develop a Custom-Built Website	7
2.3 Content Management System for Designing and Managing a Website	8
2.4 Considerations, Limitations, and Benefits of Website Designing Platforms	9
3 Task1 Part 3	10
3.1 Tools for Creating Multimedia Contents for Websites	10
3.2 Regulatory and Ethical Considerations in creating Multimedia Contents	11
References	12

Task1 Part 1

1.1 Modern Web and Mobile Development Technologies and Frameworks

Web and mobile application development technologies and frameworks are pre-build structures used intending to make the process of building faster and efficient while dealing with larger and robust applications.

Web Development:

Node.js: A JavaScript runtime built on the Chrome V8, which allows server-side execution of JavaScript and enables developers to build high-performance web applications.

React: A JavaScript library for building dynamic user interfaces on web applications using reusable UI components.

Angular: A TypeScript based framework developed by google for building dynamic web applications which allows developers to use a comprehensive set of tools and features for building large-scale applications.

Mobile Development:

Xamarin: A framework owned by Microsoft that allows developers to build cross-platform mobile apps using C#. It also provides access to application programming interfaces and code sharing across iOS, Android, and Windows platforms.

React Native: A framework based on *React* that allows developers to build cross-platform mobile applications using *JavaScript* and *React*. React Native allows for code reprocess and offers a native-like user practice.

1.2 Evaluating the Impact of Common Development Technologies and Frameworks on Design, Functionality and Management

Two of the most common development technologies and frameworks include Django and Angular. These technologies have a significant impact on design, functionality, and management as:

Design: Angular offers ready-made components and UI libraries to create visually appealing and consistent user interfaces. They maintain code reusability and efficient rendering, resulting in better design. However, it requires deep knowledge over many concepts and tools, which can negatively impact beginners.

Django, on the other hand, provides model-view, which separates logic, presentation, and data layers of the application providing clear understanding. But it is less flexible, and strict over various configurations.

Functionality: Angular enables developers to create dynamic and interactive web pages, but when handling certain complex applications, it can cause performance issues also.

The back-end framework Django offers already made server-side functionality structures, which offer libraries and tools for handling routing, database integration, authentication, and other common tasks, accelerating development pace. But this can overkill tasks and add redundant complexity to small projects.

Management: In order to improve efficiency, code quality and project management, management is crucial to optimize code, and streamline development workflows. Angular helps coders to manage codebase with tools like Angular CLI, TypeScript and so on. However, the syntax used here is complicated and can cause confusion.

Django follows a principle named “don’t repeat yourself (DRY)” and encourages code reuse and modularity resulting in better coding habits. But these have a negative effect on performance and scalability while dealing with complex tasks.

Overall, common development technologies and frameworks play a vital role in determining the design, functionality, and management of software applications. They provide developers with efficient tools, patterns, and best practices, leading to improved productivity and maintainability along with various limitations.

1.3 Impact of Website Design on Search Engine Results

In today's Search Engine Optimization (SEO) world, web design practices play a key role. User experience has become a crucial search engine ranking factor. Because of that, the design of the website and how users interact with it directly affect site's rankings in the search results. [SEO Site Checkup, 2022]

- A well-structured website with clear navigation helps search engines index the site more effectively. For an instance, logical and intuitive menus and internal linking significantly improve the user experience and make it easier for search engines to understand the interactions among various pages.
- With the increasing use of mobile devices, search engines list mobile-friendly websites on top of the hierarchy. Responsive design, which adjusts the layout based on the screen size, ensures that the website is accessible and usable across different devices, positively impacting search engine rankings.
- Clear and descriptive URLs that include relevant keywords can contribute to better search engine visibility. Avoiding usage of complex or randomly generated URLs and use hyphens to separate words enhances comprehensibility.

1.4 Improving Website Ranking Using SEO

Navigation Structures: these affect important metrics like bounce rate, engagement rate, and conversion rate. Enhancing such factors not only help improving search rankings, but they also increase revenue and profit scale. To create a good navigation structure:

- Creators can keep the main menu simple limiting the number of redundant items.
- Using familiar terms like home for home page rather than using complex technical jargon
- It is crucial to keep the navigation bar easily accessible on all smaller devices.
- Adding a functional and advanced search bar for helping the user what they are looking for.

Loading Speed: website loading time has a significant effect on search engine ranking. Generally, a website should take less than a second to load and to improve this:

- CSS and JS files should be minified.
- Usage of compressed images is crucial, precisely on those in the subheadings.
- HTTP request and the use of redirects should be minimized to provide users fast browsing.

Reading Experience: A pleasant reading experience can lead to **higher on-page time**, lower bounce rate, and an increased engagement rate, which eventually translate into better search engine rankings.

- Color combination is vital element to increase readability of users. Implementing the 60-30-10 rule over a reasonable brand color, yet readable, increases users' likelihood of staying more time over the site.
- Using standard fonts such as sans serif, dividing the contents into headings/paragraphs based upon their need, using of infographics, and videos where appropriate to make it interesting can significantly help these factors.

A well-designed website ensures a better user experience, which is now a crucial factor in search engine optimization. It's important to understand that these two aspects are now intertwined, and both play an important role in ensuring the website remains at the top of the Search engine results pages. [SEO Site Checkup, 2022]

Task1 Part 2

2.1 Design Flexibility, Performance, Functionality, User Experience (UX) and User Interface (UI)

While designing a website, various factors are required to be considered to build a user-friendly and successful site. Some of these are the responsiveness, performance level, interactivity and their functionality, UX and UI, and so on.

Flexibility: this refers to the responsiveness of a site, where site is displayed effectively over a range of devices without hampering appealing UI and different functionalities. As more and more usage of mobile devices of different sizes are increasing, this factor is getting crucial. CSS media query can play this role effectively by choosing what happens when a certain resolution is met.

Performance: it involves factors such as page loading speed, efficient code, optimized images, and minimal use of external resources. By optimizing performance, websites can provide a fast and seamless browsing experience, resulting in being ranked up in the search engine results. Using compressed codes, minimized CSS and JS, avoiding duplications and junk codes can enhance site performance.

Functionality: this refers to the features and capabilities offered by a website and typically includes interactive elements, forms, search functionality, e-commerce capabilities, content management systems, and any other tools or functionalities that enable users to accomplish their goals on the website.

User Experience (UX): factors such as ease of navigation, intuitive interaction, clear information hierarchy, and efficient user flows, are generally apart of UX. UX design aims to understand users' needs and goals to deliver a website that is user-friendly, enjoyable, and united with their expectations.

User Interface (UI): UI design focuses on the visual and interactive elements of a website, including typography, colours, layouts, icons, buttons, and other graphical components. A good UI would effectively increase UX.

2.2 Tools, Techniques and Languages Used to Design and Develop a Custom-Built Website

While choosing various strategies for designing and developing website from scratch, there are several factors to be considered like budget, ease of use, community support, and project requirements. Depending upon needs and situation these tools and techniques may vary.

Structure: while building up a structure of a site HTML is the key component. While CSS is used to manipulate the visual appearance of the HTML structure. Both these in addition are the fundamental building blocks for website design. These are completely free and can be written using Notepad++. But this requires deep understanding of these languages and may often become troublesome while debugging.

Interactivity: to add interactive features like search options or moving texts, pop-ups, responsive buttons, JavaScript comes in the first place. It also offers client-side validations. It is supported by all major web browsers and is free to use. But using this also makes the website vulnerable to various scripting attacks.

Designing and Editing: during the journey of site designing, graphics and other visual effects will be required at various times, including logo, content plates, backgrounds and likewise. Sketch is widely used for creating and editing graphics, logos, and other graphical assets, and it also offer free options, while Adobe Photoshop, Illustrator requires paid subscription. However, considering free options might have limitations also.

Hosting: web hosting services like Bluehost, SiteGround, and AWS provide the infrastructure and server resources for hosting websites. But using these requires appropriate management and some options might even require purchasing. Infinity-free also provides several features for totally free of cost and provides with sub-domain names either.

Data Analysis: tools like Google Analytics and Hotjar help track website traffic, user behaviour, and performance. But these can have limitations and may occasionally generate false reports.

2.3 Content Management System for Designing and Managing a Website

CMS platforms and pre-built sites provide a more accessible option for creating websites with existing templates and features, but customization options are more limited compared to custom-built solutions.

- Users can build sites without having extensive coding knowledge. All they need to do is select, drag, and drop.
- Pre-built templates make the process of designing and developing faster.
- The sites built through these platforms usually generate full-flexible websites resulting in better UX and saving time due to the redundancy of robust coding and analysis.
- Despite the good sides, it also has some limitations as these sites offer various membership plans. Free users will have consistent advertisement of the platform and other restrictions to premium features.
- CMS platforms can be resource intensive, resulting in reduced performance levels.

Overall, CMS platforms provide a convenient and efficient way to design and manage websites, particularly for non-technical users or small to medium-sized websites. They offer flexibility, ease of use, and a range of features. However, for complex or highly customized websites, a custom-built approach is more suitable.

2.4 Considerations, Limitations, and Benefits of Website Designing Platforms

Building websites on a designing platform has its own benefits and drawbacks comparing to traditional way of building them from scratch. All though it saves time to use builders, but it has certain limitations and things to be considered.

Considerations:

- Although web-designing platforms offer high and consistent flexibility within their built-in templates, it may be challenging to achieve unique and advanced design approaches.
- Advanced functionality and scalability may not be achieved within these platforms due to certain limitations.
- Using free membership can cause one's site to have brand logo or consistent advertisements over it.

Limitations:

- These platforms offer a set of templates; constantly used by many other sites also, which can ultimately hurt brand uniqueness.
- Designed websites with a certain platform gets dependant over it and migrating it to different hosting environment may require vigorous effort and knowledge or might even be restricted by terms.

Benefits:

- These platforms are designed to make the concept of designing easy and simple even to the ones with no technical knowledge, which ultimately saves time and labour.
- These platforms handle technical supports and provide security, ensuring the built site remains stable and secure.

Task1 Part 3

3.1 Tools for Creating Multimedia Contents for Websites

Multimedia contents are type of entities that contain messages in a visually appealing way. Combination of different texts with various colour, gradient, fonts; images, audio, and video to deliver info, all are part of the digital multimedia content.

Studies have shown that the human brain processes visual information faster than text. That means internet “grazers” who only glance at a certain site’s content will absorb a great deal more of their message, and users might even get into staying longer. [DevriX, 2022]

Various tools are available for multimedia creation and some of the great ones are:

Canva: it is a online which allows users to create diagrams, infographics, presentations and other visual aids for free but with a limited storage.

Odyssey: it is a web tool that allows users to create interactive stories that have some kind of locational element to them. The tool uses CartoDB as back-end. Although, it guides user through the process, some technical knowledge could help throughout using this. [UC Berkeley, N.D]

Theme X: it is a WordPress theme that allows a prominent level of customization and multimedia integration to a WordPress site. Though various extensions, the theme offers a drag-and-drop interface for creating customized pages and this could cost relevantly. [UC Berkeley, N.D]

Infogram: an online service that allows users to create infographics using a simple drag-and-drop interface. The graphics are professional that data visualization programs would produce, but the capabilities are limited to simple datasets. The best way to use this is to let the graphic lead the data format because it will quickly unveil its limitations. [UC Berkeley, N.D]

3.2 Regulatory and Ethical Considerations in creating Multimedia Contents

Regulatory Considerations:

- Ensuring that multimedia content, such as images, videos, or music, used on the website complies with copyright laws. Obtain proper permissions or licenses for copyrighted material to avoid infringement. [U.S. Copyright Office, 2021]
- Protecting the privacy of users by complying with data protection regulations, such as the General Data Protection Regulation (GDPR) in the European Union. [EU-GDPR, 2021]

Ethical Considerations:

- Representing diverse perspectives and avoiding content that promotes discrimination, stereotypes, or hate speech. [UNESCO, 2020]
- Obtaining user consent for collecting personal information and clearly communicating how data will be used. [ICO, 2021]

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